UDCT v7.13 | Preregistration

Refined Jeans Sign-Boundary and Hessian Diagnostics in Hydrus I

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Status: prospective protocol and preparation-only package. No v7.13 galaxy simulation or new Hessian measurement has been performed. Reserved DOI is not a published preregistration.

Records and provenance

GitHub: https://github.com/wspaik5/UDCT_v7.13_Preregistration_Jeans_Boundary_Refinement
Zenodo reserved DOI: https://doi.org/10.5281/zenodo.23342971
v7.12 post-registration: https://doi.org/10.5281/zenodo.23296444
v7.12 preregistration: https://doi.org/10.5281/zenodo.23280972

Motivation and inherited evidence

The frozen v7.12 post-registration tables contain 363 fixed cases: 274 N, 48 P and 41 I. All 24 adaptive raw-sign-change brackets stopped at endpoint or midpoint numerical uncertainty margins; none met its registered coordinate precision. The v7.13 study is designed to distinguish resolvable sign boundaries from numerical non-resolution, without changing the underlying gravity law or treating a one-dimensional Jeans test as a direct 3-D Hessian measurement.

Frozen sample and primary endpoints

All 24 inherited unresolved brackets in adaptive_brackets.csv are included without selection. The primary endpoint is a per-target outcome: (i) validated strict opposite-sign bracket, (ii) unresolved numerical margin, (iii) invalid/no robust sign bracket, or (iv) failed validation gate. Any claimed critical coordinate must also meet independently preregistered width and error requirements; otherwise no exact root is reported.

Prospective numerical and Hessian procedure

1. Preserve the v7.12 physics, 24 original brackets, and four deciding configurations: default/refined grids with inner radii 0.002 and 0.0005 r_h.

2. Freeze and publish an implementation amendment defining an independent accuracy arm, Hessian estimator, tolerances, error propagation and exact stopping criteria before any v7.13 scientific execution. No such new numerical implementation is included or claimed here.

3. Evaluate every bracket in fixed B01–B24 order; verify isotropic/field consistency; preserve all attempt logs, errors and nonconvergence. A sign is robust only outside its registered uncertainty envelope.

4. Apply sign-bracket tests to the sampled normalized Jeans second moment F, not merely N/P labels. Do not assume monotonicity, uniqueness or that a bracket proves an exact root.

5. Any Hessian diagnostics must come from an explicit, converged field derivative calculation; report their relation to Jeans classifications as exploratory unless a separately frozen quantitative hypothesis exists.

Pre-execution lock and falsifiability

This record freezes the target set and reporting rules but does not lock a new resolution arm or Hessian estimator. Therefore a later public, timestamped amendment is mandatory before confirmatory runs. If it is absent, any new calculations are exploratory and cannot be described as tests of a fully prespecified v7.13 numerical protocol. All

negative, indeterminate and failed outcomes must be reported.

Interpretation limits

N is incompatibility with a specified stationary Jeans closure; P is only a positive sampled second-moment candidate; I indicates numerical uncertainty. None establishes a positive distribution function, equilibrium, stability, observational fit, universal Hessian threshold, or general exclusion of MOND.

Target inventory (all 24)

ID	Axis	Law	Fixed	Range	Prior stop
B01	beta	BASE	1.0	0.35–0.4	endpoint
B02	beta	BASE	1.2	0.4–0.45	endpoint
B03	beta	BASE	1.5	0.45–0.5	endpoint
B04	ell_pc	BASE	0.4	1.0–1.2	endpoint
B05	ell_pc	BASE	0.45	1.2–1.5	endpoint
B06	ell_pc	BASE	0.5	1.5–2.0	endpoint
B07	beta	A-f2	1.0	0.4–0.45	endpoint
B08	beta	A-f2	1.2	0.4–0.45	midpoint
B09	beta	A-f2	1.5	0.45–0.475	midpoint
B10	ell_pc	A-f2	0.45	1.2–1.5	midpoint
B11	ell_pc	A-f2	0.5	1.5–2.0	endpoint
B12	beta	A-Qs	1.0	0.35–0.4	midpoint
B13	beta	A-Qs	1.2	0.35–0.4	endpoint
B14	beta	A-Qs	1.5	0.4–0.45	midpoint
B15	beta	A-Qs	2.0	0.475–0.5	midpoint
B16	ell_pc	A-Qs	0.4	1.2–1.5	endpoint
B17	ell_pc	A-Qs	0.45	1.5–1.75	midpoint
B18	ell_pc	A-Qs	0.5	2.0–2.25	midpoint
B19	beta	A-Qd	1.0	0.35–0.4	endpoint
B20	beta	A-Qd	1.2	0.4–0.45	midpoint
B21	beta	A-Qd	1.5	0.45–0.475	midpoint
B22	ell_pc	A-Qd	0.4	1.0–1.2	endpoint
B23	ell_pc	A-Qd	0.45	1.2–1.5	midpoint
B24	ell_pc	A-Qd	0.5	1.75–2.0	midpoint

Reproducibility and provenance

The accompanying archive contains unchanged inherited scientific source files, the two CSV inputs, a standard-library preflight checker, SHA-256 inventory, README, and this protocol PDF. The source v7.12 ZIP did not contain the complete raw compressed production result or its full incremental journal. No v7.13 post-registration results are included.